INCH-POUND

MIL-DTL-83521/5A
21 March 2018
SUPERSEDING
MIL-M-83521/5
5 August 1985

DETAIL SPECIFICATION SHEET

MAINTENANCE KITS, ELECTRICAL CONNECTORS, CONTACTS AND CONNECTOR AND CABLE ACCESSORIES: MAINTENANCE KIT, GENERAL PURPOSE (AIR/NITROGEN HEAT GUN)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-83521.

Inactive for new design after 21 March 2018.

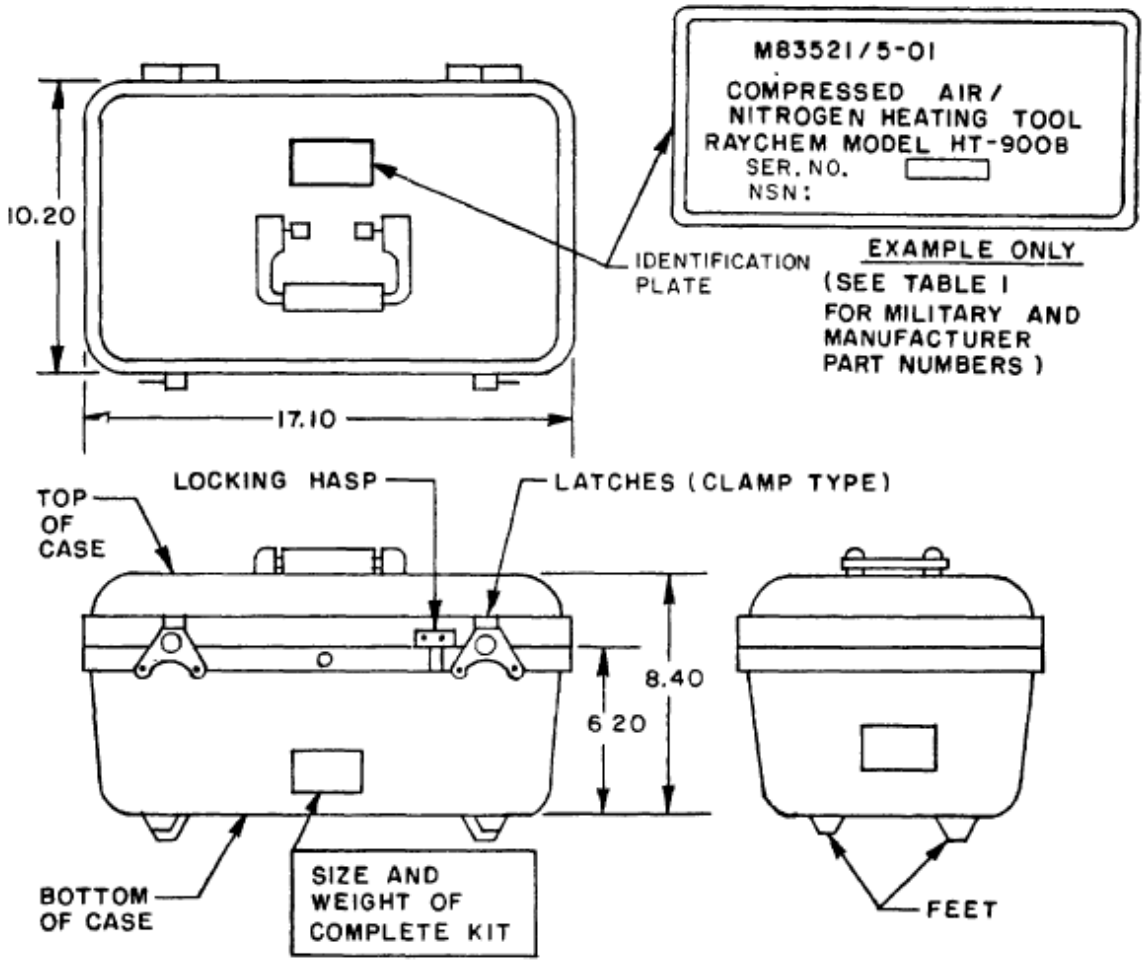


FIGURE 1. Maintenance kit, general purpose (air-nitrogen heat gun).

AMSC N/A

FSC 5935

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:33Z
Check the source to verify that this is the current version before use.

PARTS LIST:

(1) M83521/5-04	Tool-case assembly (olive drab 115 Vac)
(1) M83521/5-05	Tool-case assembly (yellow 115 Vac)
(1) M83521/5-06	Tool-case assembly (olive drab 230 Vac)
(2) Raychem part number 508744	Heating element – 115 Vac (or approved equivalent)
(1) Raychem part number 011349	Heating element – 230 Vac (or approved equivalent)
(2) Raychem part number 729711	Thermocouple (or approved equivalent).
(1) Raychem part number 491878	Termination sleeve reflector (or approved equivalent)
(1) Raychem part number 979663	Miniature termination sleeve reflector (or approved equivalent)
(1) Raychem part number 979961	Boot and tubing reflector (or approved equivalent)
(1) Raychem part number 979647	Needlepoint reflector (or approved equivalent)
(1) Raychem part number 444100	Large boot or tubing reflector (or approved equivalent)
(1) Raychem part number 369118	Filter element (or approved equivalent)
(1 set) Operating instructions	(Attached to cover)
(1 set) Maintenance manual	Ref. Air Force T.O. 32A20-3-74-2
(1) Raychem part number 980123	Heat gun/handle assembly – 115 Vac (or approved equivalent)
(1) Raychem part number 975825	Heat gun/handle assembly – 230 Vac (or approved equivalent)
(1) Raychem part number 074595	Handle assembly (or approved equivalent)
(1) Raychem part number 636183	Air hose assembly, with hardware (or approved equivalent)

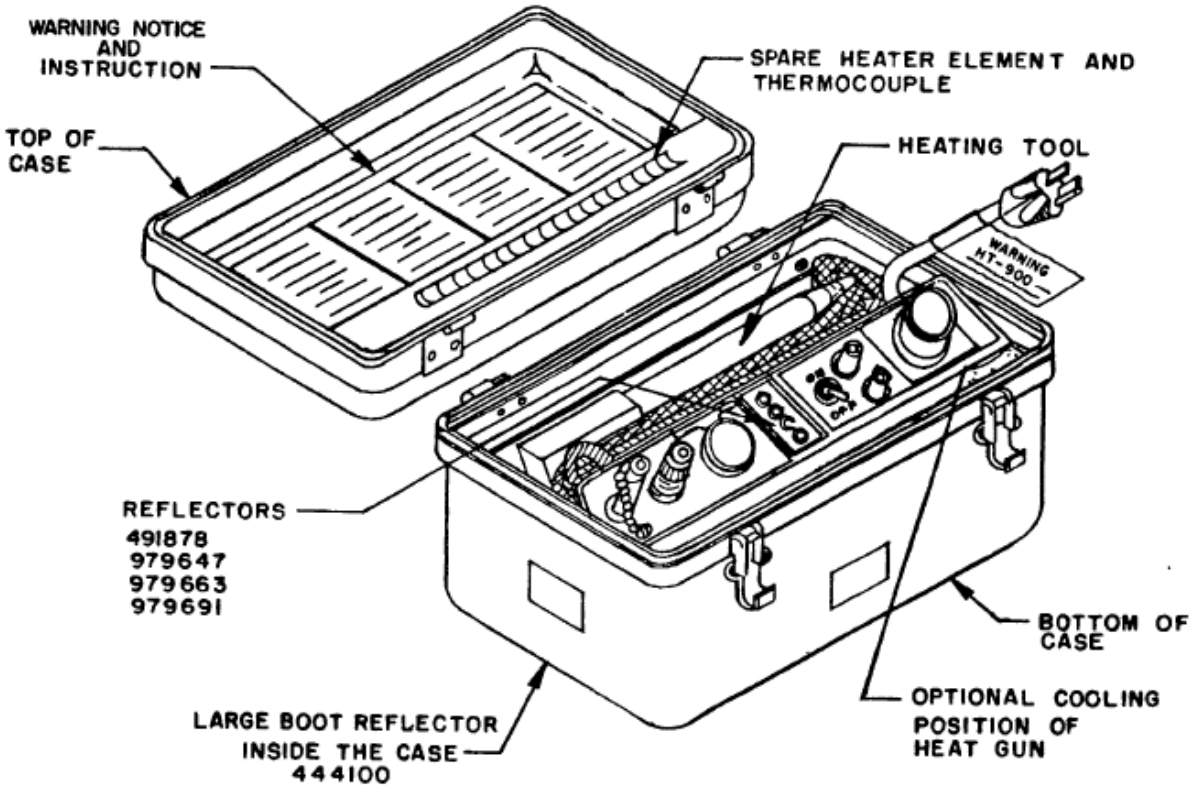
Inches	mm
6.20	157.5
8.40	213.4
10.20	259.1
17.10	454.3

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerance is ± .20 (5.1 mm).
4. Part numbers (see table I) includes case and tools.

FIGURE 1. Maintenance kit, general purpose (air-nitrogen heat gun) - Continued.

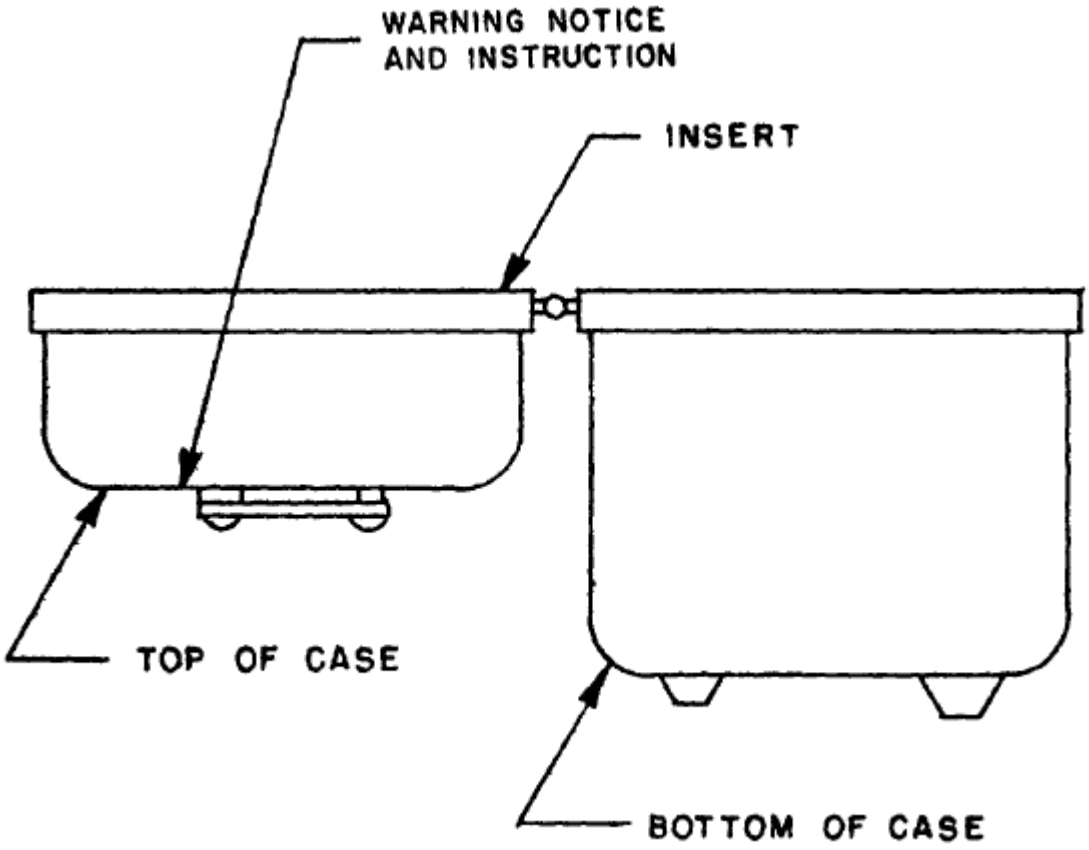
Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:33Z
Check the source to verify that this is the current version before use.



NOTE: Six digit number is Raychem Part No. “FOR EXAMPLE ONLY”.

FIGURE 1. Maintenance kit, general purpose (air-nitrogen heat gun) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:33Z
Check the source to verify that this is the current version before use.



SIDE VIEW

NOTES:

1. Part number M83521/5-04 for case only (tools not included), Air Force use.
2. Part number M83521/5-05 for case only (tools not included), Navy use.
3. Part number M83521/5-06 for case only (tools not included), Air Force Europe.
4. Tool instructions inside of lid.

FIGURE 2. Maintenance case assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:33Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

The MIL-DTL-83521/5 compressed air / nitrogen heating tool has been designed to be operated in the presence of a potentially explosive atmosphere, as defined in MIL-STD-810, without creating an explosion. The tool must be tested and must meet the requirements of MIL-STD-810, method 511.1, procedure I.

Design and configuration: See figures 1 and 2.

Warning label: An adhesive back warning label (international orange background with black lettering), appropriate to the voltage rating for 115 Vac or 230 Vac maintenance kit shall be directly behind the electrical three prong plug for M83521/5-01 and -02, and shall read on both sides of the label as follows:

HT900B (115 Vac, 50-400 Hz, 1 Ø, 7 A. The attached standard universal power plug is not approved for use in hazardous locations as defined in NFPA 70-1984, article 500. If this heating device must be used in a hazardous location as defined in NFPA 70-1984, article 500, the universal power plug must be replaced with a plug designed for use in a hazardous location.

It is the responsibility of the user to provide appropriate connection to the power source. Failure to comply with these instructions may result in fire or explosion.

For M83521/5-03, the label shall read as follows:

HT9200B (230 Vac, 50-400 Hz, 1 Ø, 3, 5 A. The attached standard universal power plug is not approved for use in hazardous locations as defined in NFPA 70-1984, article 500. If this heating device must be used in a hazardous location as defined in NFPA 70-1984, article 500, the universal power plug must be replaced with a plug designed for use in a hazardous location.

It is the responsibility of the user to provide appropriate connection to the power source. Failure to comply with these instructions may result in fire or explosion.

Materials:

Case (category 2): Fiberglass with an aluminum frame; color in accordance with SAE-AMS-STD-595 (see table II).

Lining: Closed-cell neoprene in accordance with ASTM-D1056.

Insert: Polyethylene foam in accordance with A-A-59135 or A-A59136.

Clasps, handles, hinges, and locking hasp: Suitably plated to resist corrosion.

Nameplate: In accordance with MIL-DTL-19834.

Rubber feet: On bottom of case, see figure 1.

Instruction sheets: Warning statement as appropriate (115 Vac, 230 Vac units) and operating instructions shall be printed on the inside of the top of the case. Lettering shall be legible and protected with transparent coating. Color of warning statement shall be in legible red lettering and shall precede the operating instructions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:33Z
Check the source to verify that this is the current version before use.

Heating tool: Compressed air nitrogen heating tool (Raychem HT-900B, HT-920B, or approved equivalent). The heating tool shall consist of a control module and a heat gun unit connected together by a hos assembly containing air/nitrogen lines and electrical wires. It shall operate electrically as specified in table I, and with external compressed air or nitrogen at 80-200 lb-ft / in^2. It shall produce heated air / nitrogen at a temperature of 480°C (900°F $\pm$20°F) at maximum setting. The heating tool control module shall be attached to the lower case half with an instant release latch hinge.

Heat gun: The heat gun shall be equipped with a pneumatic control device to allow the user to switch the electric power “ON” or “OFF” at the heating tool.

Reflectors: Five heat reflectors shall be stowed conveniently on the control panel of the heating tool.

Hose assembly: The hose assembly containing the air/nitrogen lines and electrical wires shall withstand 500 cycles of flexing.

Pressure switch: ITT Neo-dyne absolute pressure switch, Model No. 500P0044 or functional equivalent.

Filter-regulator: C.A. Norgren 807-134-A1KA filter regulator or equivalent.

Feedback controller: Watlow part no. 1A10-11 controller or equivalent.

Note: All electrical connections must be potted or have conformal coating applied.

Part numbers:

M83521/5-01 - Tool Kit, General Purpose, (air / nitrogen, heat gun) Air Force use.
M83521/5-02 - Tool Kit, General Purpose (air / nitrogen, heat gun) Navy use.
M83521/5-03 - Tool Kit, General Purpose, 230 Vac (air / nitrogen, heat gun) Air Force use.
M83521/5-04 - Tool Case Assembly, (color, olive drab) 115 Vac.
M83521/5-05 - Tool Case Assembly, (color, yellow) 115 Vac.
M83521/5-06 - Tool Case Assembly, (color, olive drab) 230 Vac.

TABLE I. Maintenance kit characteristics.

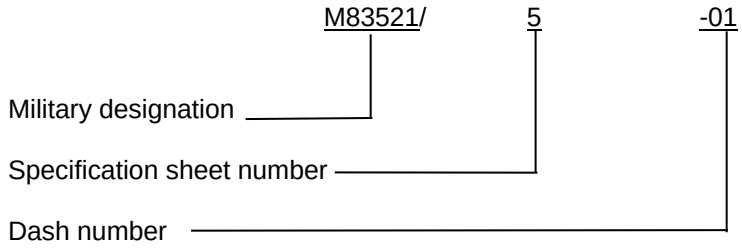
Maintenance kit military part number	Voltage	Frequency	Phase	Power (amperes)	Case color	Superseded part no.
M83521/5-01	115 Vac	50-400 Hz	Single	6	Olive drab	M83507/14-01A
M83521/5-02	115 Vac			6	Yellow	None
M83521/5-03	230 Vac			2.75	Olive drab	None

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:33Z
Check the source to verify that this is the current version before use.

TABLE II. Maintenance case characteristics.

Maintenance case military part number	Voltage	Case color	Use	Superseded part no.
M83521/5-04	115 Vac	Olive drab	Air Force	M83507/14-02A
M83521/5-05	115 Vac	Yellow	Navy	None
M83521/5-06	230 Vac	Olive drab	Air Force	None

Part or Identifying Number (PIN) example:



Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-83521, this document references the following:

MIL-DTL-19834
MIL-STD-810
A-A-59135
A-A-59136
SAE-AMS-STD-595
ASTM-D1056
NFPA70
AF T.O. 32A20-3-74-2

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2018-062)

Review activities:
Army - AR, AT, GL, MI
Navy - EC, MC, OS, SH
Air Force - 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:33Z
Check the source to verify that this is the current version before use.